



VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

25PY901

Engineering Physics

Year 1 Semester 2 Section 24 (ECE + ECEVLSI)

Course instructor: Dr. Sreekar Guddeti

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Module Bank

Module 2

Instructions:

1. Answer all questions.
2. BT stands for Blooms Taxonomy.

PART-A

- (1) Quantum Free Electron Theory (QFET) [10 M]
- (a) Compare and contrast the postulates of classical free electron theory with that of quantum free electron theory. [BT 1][6 M]
 - (b) What are the merits of quantum free electron theory? [BT 1][2 M]
 - (c) What are the demerits of quantum free electron theory? [BT 1][2 M]
- (2) Fermi-Dirac Distribution [10 M]
- (a) Plot the probability of occupancy of energy level in the case of Maxwell-Boltzmann distribution and Fermi-Dirac distribution at [BT 2][2 M]
 - (i) $T = 0\text{ K}$
 - (ii) $T \neq 0\text{ K}$
 - (b) Five free electrons exist in a three-dimensional infinite potential well with all three widths equal to $a = 12\text{ \AA}$. [BT 3][4 M]
 - (i) Determine the Fermi energy level at $T = 0\text{ K}$.
 - (ii) Repeat part (i) for 13 electrons.
 - (c) Fermi level of a metal is 6.25 eV and electrons in this material follow the Fermi-Dirac distribution. Calculate the temperature at which there is 1% probability that a state 0.30 eV below the Fermi energy level will not contain an electron. [BT 3][4 M]
- (3) Electronic specific heat of solids [10 M]
- (a) Define specific heat of a material. What is its SI unit? [BT 1][2 M]
 - (b) Specific heat of mercury $^{200}_{80}\text{Hg}$ is $0.14\text{ J g}^{-1}\text{ K}^{-1}$. Report its specific heat in SI units. [BT 1][2 M]
 - (c) Derive the expression for specific heat capacity from classical free electron theory and quantum free electron theory. [BT 4][6 M]
- (4) Density of states [10 M]
- (a) What is the meaning of density of states function? What is its SI unit? [BT 1][2 M]
 - (b) If the density of states of a material is a constant $Z(E) = k$ and the electron density is n , [BT 3][4 M]
 - (i) What is the Fermi level at $T = 0\text{ K}$?
 - (ii) What is the Fermi level at $T \neq 0\text{ K}$?
 - (c) Sketch the density of states function [BT 2][4 M]
 - (i) for the conduction band of an intrinsic semiconductor.
 - (ii) Repeat part (i) for the valence band.

(5) E vs k diagram [10 M]

- Sketch the E vs k diagram for a free electron. [BT 1][1 M]
- Sketch the E vs k diagram for an electron in a periodic potential. [BT 1][1 M]
- Consider an intrinsic semiconductor with all electrons occupying states in the lowest valence band at $T = 0$ K. Plot the E vs k showing occupancy of levels at both $T = 0$ K and $T \neq 0$ K. [BT 2][2 M]
- Two possible conduction bands are shown in the E vs k diagram given in Figure. 1. State which band will result in the heavier electron effective mass; state why [BT 2][2 M]

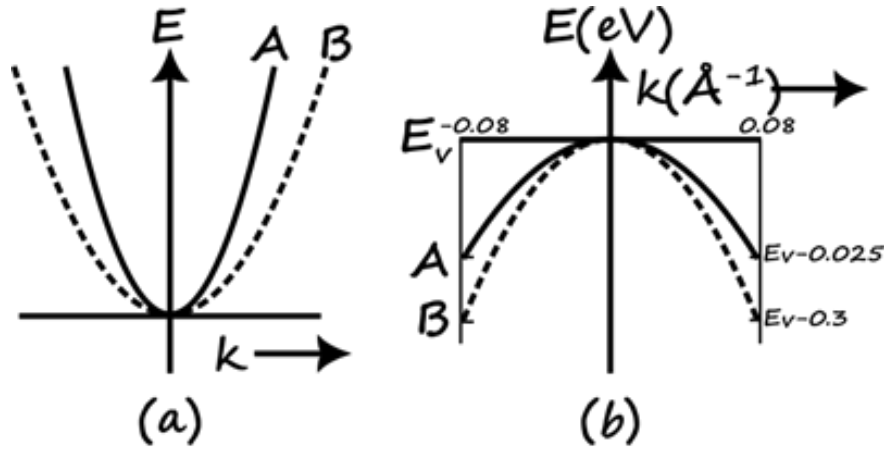


Figure 1: (a) Conduction and (b) valence bands of materials A and B.

Figure 1:

(6) Fermi level in semiconductors [10 M]

- Sketch the energy band diagram showing the density of states, Fermi-Dirac probability function and areas representing electron and hole concentration for an intrinsic semiconductor at $T \neq 0$ K. Assume electron effective mass is equal to hole effective mass. [BT 1, 2][3 M]
- Calculate position of Fermi level with respect to center of the bandgap in silicon for
 - $T = 0$ K
 - $T = 300$ K

The electron effective mass is $1.08m_0$ and hole effective mass is $0.56m_0$.

- Draw, compare, and contrast the sketches of the energy band diagram for p-type semiconductor at $T = 0$ K and $T = 300$ K. [BT 3][2 M]
- Silicon atoms, at a concentration of $7 \times 10^{15} \text{ cm}^{-3}$, are added to GaAs. Assume that the silicon atoms act as fully ionized dopant atoms and that 5% of the dopants replace Ga atoms and 95% replace As atoms. [BT 2][3 M]

- (i) Determine the donor and acceptor concentrations.
 - (ii) Is the material n type or p type?
- (7) p-n junction diode [10 M]
 - (a) Write the equation and sketch the $I - V$ characteristics of pn junction diode.[BT 1][2 M]
 - (b) Define reverse saturation current of pn junction diode. Derive it from the $I - V$ characteristics. [BT 1][2 M]
 - (c) A silicon diode can be used to measure temperature by operating the diode at a fixed forward-bias current. The forward bias voltage is then a function of temperature. Assume that the reverse saturation current is proportional to $\exp(-E_g/k_B T)$. At $T = 300\text{ K}$, the diode voltage is found to be 0.60 V . Determine the diode voltage at $T = 310\text{ K}$. What is the percentage change in voltage? [BT 4][3 M]
 - (d) The pn junction diode can also be used as a temperature sensor by operating the diode at a fixed reverse-bias voltage. What is the percentage change in current for the same temperature rise? [BT 4][2 M]
 - (e) Which mode is more sensitive, the constant forward current mode or constant reverse voltage mode? [BT 2][1 M]
- (8) Solar cell [10 M]
 - (a) Sketch the $I - V$ characteristics of pn junction solar cell. [BT 1][1 M]
 - (b) Define short circuit current and open circuit voltage. [BT 1][2 M]
 - (c) A silicon solar cell under standard test conditions has short circuit current density of 35 mA cm^{-2} , and open circuit voltage of 0.60 V . The fill factor for the material is 0.80 . If the incident power is 100 mW cm^{-2} , calculate the efficiency of the cell. [BT 3][4 M]
 - (d) The energy band gap of a semiconductor is 3 eV . Is this material useful for solar cell applications? Give reasons. [BT 4][3 M]
- (9) Direct and indirect band gap semiconductors [10 M]
 - (a) Sketch the E vs k diagram for [BT 1][2 M]
 - (i) a direct bandgap semiconductor, and
 - (ii) an indirect bandgap semiconductor.
 - (b) Why is direct band gap semiconductor useful for solar cell application? Crystalline silicon is indirect band gap semiconductor. What is the process employed in industry to “convert” it into a direct band gap semiconductor for solar cell application? [BT 2][2 M]
 - (c) c) Which material is used in blue LED? What is the bandgap of the material? [BT 3][3 M]

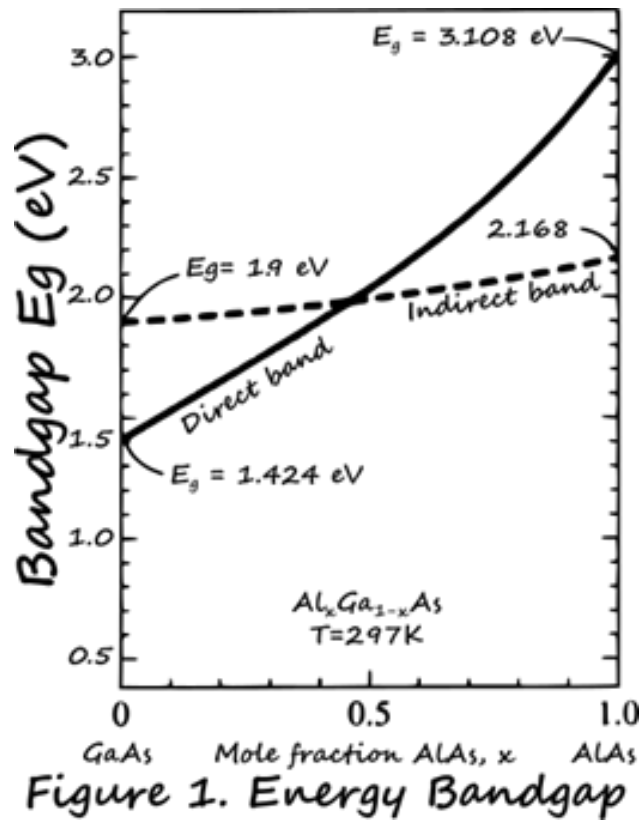


Figure 2:

- (d) $Al_xGa_{1-x}As$ is a ternary semiconductor and the energy band gap variation as a function of mole fraction x of AlAs is shown in Figure. 2. Which composition can be used as material for blue LED application? [BT 4][3 M]

(10) LED [10 M]

- (a) What are the materials used for red, green, and blue LEDs? [BT 1][2 M]
- (b) What are the mechanisms that decrease the efficiency of an LED? [BT 1][3 M]
- (c) The energy band gap variation as a function of mole fraction x of $GaAs_{1-x}P_x$ is shown in Figure. 3. What is the range of colours that can be emitted if used for LED application? Report in wavelength. [BT 3][3 M]
- (d) Calculate the critical angle at GaAs - air interface. The refractive index of GaAs is 3.8 at $\lambda = 0.70 \mu m$. [BT 3][2 M]

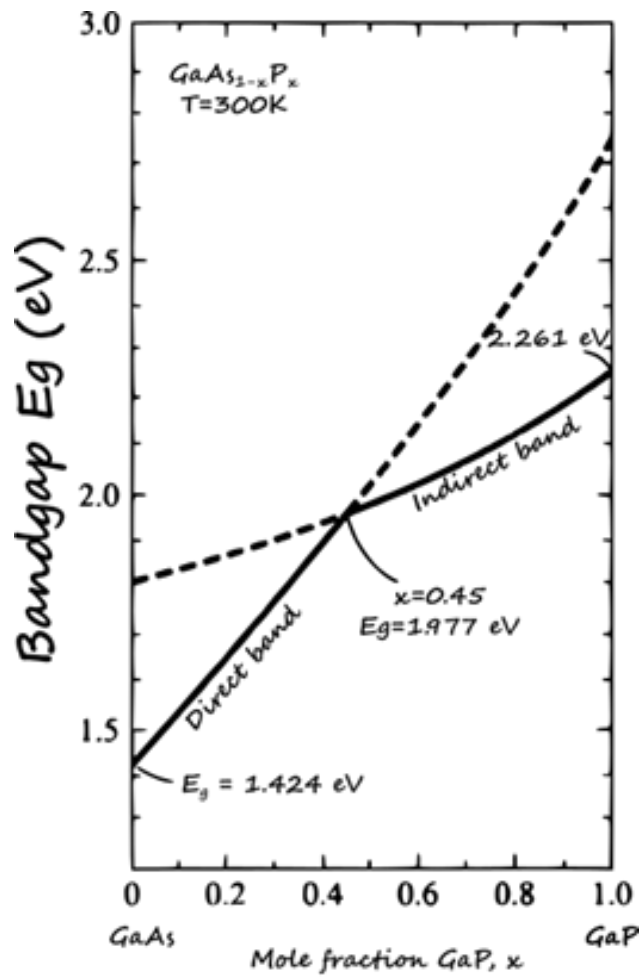


Figure 1. Energy band gap

Figure 3: